Data Setup & Initial EDA Report

Week 1 — Opportunity Data & Learner Data (Excelerate)

Prepared by: Emmanuel Erioluwa Olowodu

1. Database Setup

The working environment was built on a Chromebook, which cannot run PostgreSQL locally without significant workarounds (e.g. enabling a Linux container). To avoid this limitation, Supabase was used — a hosted PostgreSQL platform that runs entirely in the browser and provided the PostgreSQL database required by the brief without any local installation.

Database Creation

A new Supabase project was created, which provisions a dedicated PostgreSQL instance. This instance served as the working database for both datasets.

Dataset Upload Process

Both the Opportunity Data and Learner Data CSV files were uploaded via Supabase's Table Editor "Import data from CSV" feature, which builds a new table directly from the CSV's own column headers. Several issues were encountered and resolved during this process:

Issue Encountered	Resolution
CSV headers did not match an existing table's columns ("incompatible headers" error)	Deleted the mismatched table and re-imported using the CSV-first table creation flow, so columns were built directly from the file's own headers.
"Duplicate key value violates unique constraint" on the pk column	The pk column was not actually unique (most rows shared the value "Opportunity#"). The Primary Key setting was removed from pk prior to import.
Browser refreshed mid-import, leaving a partial/incomplete set of rows	The affected table was truncated (emptied) and the CSV was re-imported from scratch.
Learner Data CSV rejected: "ensure all your columns are named"	The source header row had roughly 212 blank column headers. A spreadsheet formula was used to auto-fill blank headers

	with placeholder names (col_45, col_46, etc.) before re-exporting and re-uploading.

Connection Method

Supabase (PostgreSQL) was selected as the connection method, in line with the brief's suggested options. A direct connection was configured and successfully tested in Looker Studio using the project's host, port, database name, username, and password — both tables (OpportunityData, LearnerData) were visible and selectable as data sources, confirming the Supabase-to-Looker-Studio pipeline works correctly.

Confirmation of Successful Load

Dataset	CSV Row Count	Database Row Count	Status
Opportunity Data	5,733	5,733	Matched — verified via SELECT COUNT(*)
Learner Data	1,979 (after header fix)	1,979	Matched — verified via SELECT COUNT(*)

2. Dataset Overview

Opportunity Data

- 5,733 records, 33 columns, all imported as text data type
- Key fields identified for future analysis: category, fee, currency_type, duration, duration_type, location, is_archived, is_auto_approve, created_at, modified_at, last_date_to_apply
- opportunity_id confirmed unique across all records — no duplicates found
- Nine columns (Reward, CareerAddOn, Cohort, Eligibility, NotStartedTransaction, DropoutTransaction, Panellist, Testimonial, tracking_questions) contain embedded JSON reference structures rather than plain values, consistent with a NoSQL/DynamoDB-style source export

Learner Data

- 1,979 records, 256 columns, all imported as text data type

- Only 44 columns carry meaningful names; the remaining 212 (83%) are unnamed placeholders (col_45–col_256) due to a broken header row in the source export
- Key fields identified for future analysis: status, application_id, apply_date, created_at, completion_date, dropped_out_date, withdraw_date, from_where_did_you_hear_about_us, fee, amount_to_be_paid, payment_status
- Date fields use two different formats: apply_date is ISO 8601 text; created_at and other date fields are millisecond-epoch timestamps stored as text

Initial Observations on Structure

Both datasets appear to originate from the same underlying system (shared JSON reference patterns, shared sk/pk-style fields), suggesting they could in principle be related — a question explored in greater depth as part of the Week 2 integration work. All fields in both tables were imported as text, meaning no numeric or date analysis is possible until explicit type conversion is performed.

3. Exploratory Analysis

SQL queries using SELECT, COUNT, GROUP BY, and information_schema were used to explore structure, distributions, and data quality in both tables.

3.1 Opportunity Categories

```
SELECT category, COUNT(*) AS totalFROM opportunity_dataGROUP BY categoryORDER BY total DESC;
```

Result: Internship (1,591), Career (1,088), Competition (734), Course (632), Event (622), Engagement (459) — six distinct categories, fairly evenly represented.

3.2 Duplicate Check on opportunity_id

```
SELECT opportunity_id, COUNT(*)FROM opportunity_dataGROUP BY opportunity_idHAVING COUNT(*) > 1;
```

Result: no rows returned — opportunity_id is a genuinely unique identifier.

3.3 Missing Value Check

```
SELECT COUNT(*) AS total_rows,  COUNT(*) FILTER (WHERE category IS NULL) AS missing_category,  COUNT(*) FILTER (WHERE "Reward" IS NULL) AS missing_reward,  COUNT(*) FILTER (WHERE name IS NULL) AS missing_name,  COUNT(*) FILTER (WHERE fee IS NULL) AS missing_feeFROM opportunity_data;
```

Result: of 5,733 records, only 1 missing Reward and 1 missing name — otherwise complete.

3.4 Learner Application Status Distribution

```
SELECT status, COUNT(*)FROM learner_dataGROUP BY statusORDER BY COUNT(*) DESC;
```

Result: status is stored as numeric codes (e.g. 1070, 1010, 1120) rather than readable labels, with 643 null values — the single largest group. Early trend: a wide spread across at least 13 distinct codes suggests a multi-stage application pipeline.

3.5 Learner Completion Funnel

```
SELECT COUNT(*) AS total, COUNT(completion_date) AS completed, COUNT(dropped_out_date) AS dropped_out, COUNT(withdraw_date) AS withdrawn, COUNT(not_started_date) AS not_startedFROM learner_data;
```

Result: of 1,979 records, only 162 (~8%) show a completion date, while 55 dropped out and 55 withdrew. Notable pattern: 1,642 records (83%) have none of these four lifecycle dates filled, indicating most learner records lack a clear completion status.

3.6 Lead Source Distribution

```
SELECT from_where_did_you_hear_about_us, COUNT(*)FROM learner_dataGROUP BY from_where_did_you_hear_about_usORDER BY COUNT(*) DESCLIMIT 15;
```

Result: "Social Media" (511) and "Illinois Institute of Technology" (313) are the top two named sources, though several entries represent the same source written inconsistently (addressed in Section 4).

4. Data Quality Assessment

4.1 Missing Values

- Opportunity Data: minimal — 1 missing Reward, 1 missing name out of 5,733 records.
- Learner Data: substantial in places — 643 null status values, 658 null lead-source values, 644 null payment_status values, and 1,734 of 1,979 records with no application_id at all.

Effect on future analysis: Opportunity Data can largely be analysed as-is; Learner Data's high null rates in status and application_id mean funnel and identifier-based analysis will understate true activity unless these gaps are accounted for.

4.2 Duplicate Records

- opportunity_id (Opportunity Data): confirmed unique, no duplicates.
- application_id (Learner Data): only 85 distinct values exist across the whole column, with heavy duplication among populated values (e.g. "12345" appears 15 times) — this field cannot currently serve as a unique record identifier.

Effect on future analysis: application_id cannot be used for deduplication or for joining Learner Data to other tables without significant cleanup.

4.3 Inconsistent Formatting

- location (Opportunity Data): same values written inconsistently — "Virtual"/"virtual"/"vitrual", "Work From Home"/"WFM"/"wfm", and "null"/"Null"/"NULL" stored as literal text rather than true empty values (1,442 rows combined).
- currency_type (Opportunity Data): "EUR" and "EURO" recorded as separate categories for the same currency.
- duration_type (Opportunity Data): singular/plural inconsistency (day/days, week/weeks, month/months, hour/hours, minute/minutes, year/years), plus a typo ("yearssss") and a truncated value ("da").
- from_where_did_you_hear_about_us (Learner Data): "Social Media"/"social Media", "My Friend Or Relative"/"My Friend/Relative", "Illinois Institute of Technology"/"Illinois Tech", and "Others"/"Other" all represent duplicate categories.

Effect on future analysis: any count or chart built directly on these fields would currently split a single real category across multiple labels, understating its true total and misrepresenting distribution charts planned for Week 3.

4.4 Unusual or Unexpected Values

- 346 initial candidate rows (later confirmed as 6 genuinely corrupted rows) in Opportunity Data contained JSON fragments (e.g. %22label%22: %22Yemen%22) sitting in fields like fee, currency_type, and is_archived — evidence of a column-shift error during the original CSV export.
- Lorem Ipsum placeholder text (e.g. "Quia ad ipsum quos") found in the location field — indicating some records in Opportunity Data are dummy/test data rather than real records.
- Nine Opportunity Data columns (Reward, Cohort, Eligibility, etc.) contain embedded JSON reference codes rather than plain analysable values.

- application_id in Learner Data contains obvious test entries ("testing", "12345", "APP12345") mixed in with genuine data.
- All date fields in both tables were stored as either millisecond-epoch numbers or ISO text rather than a native date type, requiring conversion before any time-based analysis.

Effect on future analysis: these issues do not prevent exploration but must be addressed before reliable metrics or dashboards can be built — this is the basis for the Week 2 Data Cleaning & Validation work.

Summary

Both datasets were successfully loaded into a working PostgreSQL (Supabase) environment, verified for completeness, and connected to Looker Studio. Initial exploration surfaced genuine, well-evidenced data quality issues — inconsistent categorical formatting, embedded JSON reference fields, an unreliable application identifier, a large block of unnamed columns, and a small number of corrupted rows — all of which form the documented starting point for Week 2's cleaning and validation work.